## EXP 2: Numerical Simulation and Performance Analysis of Microwave Transmission Lines and Two-Port Networks Using Scattering Parameters

### *Objectives of this Experiment:*

1. To model a 50 Ω microwave two-port transmission line over the 8–12 GHz frequency range.

2. To determine characteristic impedance, propagation constant and transmission characteristics.

3. To obtain and analyze S11, S21, S12 and S22 over the specified frequency sweep.

4. To calculate return loss, insertion loss and transmission efficiency.

5. To identify the optimum frequency and compare its performance with 10 GHz.

### *CODE:*

```
clc;
clear;
close all;

%% EXPERIMENT 2: MICROWAVE TWO-PORT NETWORK

% 50 Ohm transmission line, 8-12 GHz frequency sweep

%% 1. INPUT PARAMETERS

c = 3e8;

Z0 = 50;

L = 50e-3;

f0 = 10e9;

f = 8e9:0.01e9:12e9;

fGHz = f/1e9;

%% 2. TRANSMISSION LINE MODEL

% Illustrative model: the experiment sheet does not specify
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% the physical line type, geometry or material properties.

Zc = 50 + 5*((f-f0)/(2e9)).^2;

alpha_dB_per_m = 0.5 + ...
    0.5*((f-f0)/(2e9)).^2;

alpha = alpha_dB_per_m*log(10)/20;

beta = 2*pi*f/c;

gamma = alpha + 1j*beta;

%% 3. ABCD PARAMETERS

A = cosh(gamma*L);

B = Zc.*sinh(gamma*L);

C = sinh(gamma*L)./Zc;

D = A;

den = A + B/Z0 + C*Z0 + D;

%% 4. S-PARAMETERS

S11 = (A + B/Z0 - C*Z0 - D)./den;

S21 = 2./den;

S12 = 2*(A.*D-B.*C)./den;

S22 = (-A + B/Z0 - C*Z0 + D)./den;

S11_dB = 20*log10(max(abs(S11),1e-12));

S21_dB = 20*log10(max(abs(S21),1e-12));

S12_dB = 20*log10(max(abs(S12),1e-12));

S22_dB = 20*log10(max(abs(S22),1e-12));

%% 5. PERFORMANCE PARAMETERS

RL = -20*log10(max(abs(S11),1e-12));

IL = -20*log10(max(abs(S21),1e-12));

eta = abs(S21).^2*100;

%% 6. DISPLAY RESULTS AT 10 GHz

[~,idx10] = min(abs(f-f0));
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```
fprintf('\\nMICROWAVE ENGINEERING - EXPERIMENT 2\\n');

fprintf('-----------------------------------\\n');

fprintf('Operating frequency = %.2f GHz\\n',fGHz(idx10));

fprintf('Reference impedance = %.2f Ohms\\n',Z0);

fprintf('Line length = %.2f mm\\n',L*1000);

fprintf('Characteristic impedance = %.3f Ohms\\n',Zc(idx10));

fprintf('Propagation constant beta = %.3f rad/m\\n',beta(idx10));

fprintf('S11 = %.4f dB\\n',S11_dB(idx10));

fprintf('S21 = %.4f dB\\n',S21_dB(idx10));

fprintf('S12 = %.4f dB\\n',S12_dB(idx10));

fprintf('S22 = %.4f dB\\n',S22_dB(idx10));

fprintf('Return Loss = %.4f dB\\n',RL(idx10));

fprintf('Insertion Loss = %.4f dB\\n',IL(idx10));

fprintf('Transmission Efficiency = %.4f %%\\n',eta(idx10));

%% 7. OPTIMUM FREQUENCY

[~,idxMinS11] = min(abs(S11));

[~,idxMaxS21] = max(abs(S21));

fprintf('\\nOPTIMUM FREQUENCY RESULTS\\n');

fprintf('Minimum |S11| frequency = %.2f GHz\\n',fGHz(idxMinS11));

fprintf('S11 at minimum = %.4f dB\\n',S11_dB(idxMinS11));

fprintf('S21 at minimum = %.4f dB\\n',S21_dB(idxMinS11));

fprintf('Return Loss = %.4f dB\\n',RL(idxMinS11));

fprintf('Insertion Loss = %.4f dB\\n',IL(idxMinS11));

fprintf('Maximum |S21| frequency = %.2f GHz\\n',fGHz(idxMaxS21));

fprintf('S11 at maximum transmission = %.4f dB\\n',S11_dB(idxMaxS21));

fprintf('S21 at maximum transmission = %.4f dB\\n',S21_dB(idxMaxS21));

fprintf('Return Loss = %.4f dB\\n',RL(idxMaxS21));
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```
fprintf('Insertion Loss = %.4f dB\\n',IL(idxMaxS21));

fprintf('Efficiency = %.4f %%\\n',eta(idxMaxS21));

%% 8. GRAPH 1 TO GRAPH 9 USING SUBPLOT

figure('Color','w','Position',[50 50 1400 850]);

%% GRAPH 1: INPUT REFLECTION S11

subplot(3,3,1);

plot(fGHz,S11_dB,'b','LineWidth',1.5);

xlabel('Frequency (GHz)');

ylabel('|S11| (dB)');

title('Input Reflection S11');

grid on;

xlim([8 12]);

%% GRAPH 2: FORWARD TRANSMISSION S21

subplot(3,3,2);

plot(fGHz,S21_dB,'r','LineWidth',1.5);

xlabel('Frequency (GHz)');

ylabel('|S21| (dB)');

title('Forward Transmission S21');

grid on;

xlim([8 12]);

%% GRAPH 3: REVERSE TRANSMISSION S12

subplot(3,3,3);

plot(fGHz,S12_dB,'m','LineWidth',1.5);

xlabel('Frequency (GHz)');

ylabel('|S12| (dB)');

title('Reverse Transmission S12');

grid on;
```

```
xlim([8 12]);

%% GRAPH 4: OUTPUT REFLECTION S22

subplot(3,3,4);

plot(fGHz,S22_dB,'k','LineWidth',1.5);

hold on;

plot(fGHz(idx10),S22_dB(idx10),'ko', ...
    'MarkerFaceColor','k','MarkerSize',5);

xlabel('Frequency (GHz)');

ylabel('|S22| (dB)');

title('Output Reflection S22');

grid on;

xlim([8 12]);

%% GRAPH 5: RETURN LOSS

subplot(3,3,5);

plot(fGHz,RL,'b','LineWidth',1.5);

hold on;

plot(fGHz(idx10),RL(idx10),'bo', ...
    'MarkerFaceColor','b','MarkerSize',5);

xlabel('Frequency (GHz)');

ylabel('Return Loss (dB)');

title('Return Loss');

grid on;

xlim([8 12]);

%% GRAPH 6: INSERTION LOSS

subplot(3,3,6);

plot(fGHz,IL,'r','LineWidth',1.5);

xlabel('Frequency (GHz)');

ylabel('Insertion Loss (dB)');
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```
title('Insertion Loss');

grid on;

xlim([8 12]);

%% GRAPH 7: TRANSMISSION EFFICIENCY

subplot(3,3,7);

plot(fGHz,eta,'g','LineWidth',1.5);

xlabel('Frequency (GHz)');

ylabel('Efficiency (%)');

title('Transmission Efficiency');

grid on;

xlim([8 12]);

%% GRAPH 8: PROPAGATION CONSTANT

subplot(3,3,8);

plot(fGHz,beta,'Color',[0.5 0.2 0.7], ...
    'LineWidth',1.5);

xlabel('Frequency (GHz)');

ylabel('\\beta (rad/m)');

title('Propagation Constant');

grid on;

xlim([8 12]);

%% GRAPH 9: CHARACTERISTIC IMPEDANCE

subplot(3,3,9);

plot(fGHz,Zc,'Color',[0.85 0.4 0.1], ...
    'LineWidth',1.5);

hold on;

yline(Z0,'--k','50 Ohm Reference');

xlabel('Frequency (GHz)');

ylabel('Zc (Ohms)');
```

```
title('Characteristic Impedance');

grid on;

xlim([8 12]);

sgtitle('Microwave Two-Port Network: 8-12 GHz');

%% 9. PERCENTAGE DEVIATION

% Compare with the ideal matched reference:
% S21 ideal = 0 dB, insertion loss = 0 dB, efficiency = 100%.
% Ideal S11 and S22 are zero, so their dB values are -Inf.
% Percentage deviation is not meaningful for a zero reference.

fprintf('\\nIDEAL MATCHED REFERENCE\\n');

fprintf('S11 = 0 reflection (ideally -Inf dB)\\n');

fprintf('S22 = 0 reflection (ideally -Inf dB)\\n');

fprintf('S21 = 0 dB; S12 = 0 dB\\n');

fprintf('Return Loss = Infinite\\n');

fprintf('Insertion Loss = 0 dB\\n');

fprintf('Efficiency = 100 %%\\n');

%% 10. CONCLUSION

fprintf('\\nCONCLUSION\\n');

fprintf('S-parameters were evaluated over 8-12 GHz.\\n');

fprintf('Return loss, insertion loss and efficiency were calculated.\\n');

fprintf('The optimum frequencies were identified from the sweep.\\n');

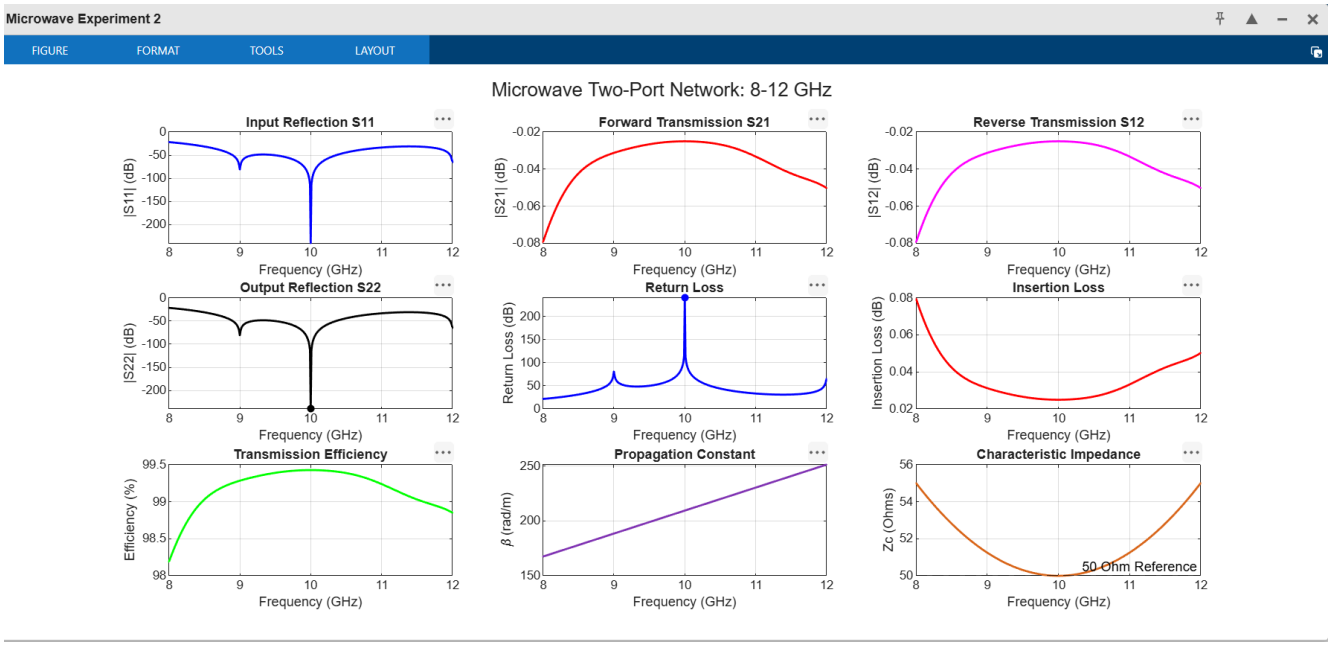
fprintf('This is an illustrative model, not validated EM-solver data.\\n');

%% DISPLAY SELECTED SWEEP RESULTS

Results = table(fGHz(:),S11_dB(:),S21_dB(:), ...
    S12_dB(:),S22_dB(:),RL(:),IL(:),eta(:), ...
    'VariableNames',{'Frequency_GHz','S11_dB','S21_dB', ...
    'S12_dB','S22_dB','ReturnLoss_dB', ...
    'InsertionLoss_dB','Efficiency_percent'});

disp(Results(1:20:end,:));
```

# OUTPUT:


Microwave Experiment 2
FIGURE
FORMAT
TOOLS
LAYOUT
Microwave Two-Port Network: 8-12 GHz
Input Reflection S11
|S11| (dB)
Forward Transmission S21
|S21| (dB)
Reverse Transmission S12
|S12| (dB)
Output Reflection S22
|S22| (dB)
Return Loss
Return Loss (dB)
Insertion Loss
Insertion Loss (dB)
Transmission Efficiency
Efficiency (%)
Propagation Constant
β (rad/m)
Characteristic Impedance
Zc (Ohms)
50 Ohm Reference
Frequency (GHz)